The logo is a square box containing the text "amrita_logo.png".

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**Secure Blockchain-Based Financial Transaction System
Using Cryptographic Primitives and Smart Contracts**

*A Project Report submitted in partial fulfillment
of the requirements for the degree of*

**Bachelor of Technology
in Computer Science and Engineering**

Submitted by: Dharaneswara Reddy

Register No: [Your Register Number]

Under the guidance of: [Guide Name], [Designation]

Academic Year: 2025 – 2026

Semester: VI

CERTIFICATE

This is to certify that the project report entitled “**Secure Blockchain-Based Financial Transaction System Using Cryptographic Primitives and Smart Contracts**” submitted by **Dharaneswara Reddy** (Register No: [Your Register Number]) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at Amrita Vishwa Vidyapeetham, Coimbatore, is a record of bonafide work carried out by the student under our supervision during the academic year 2025–2026.

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ABSTRACT

Financial transactions in centralized banking systems are managed by trusted intermediaries such as banks and payment processors. These systems are susceptible to fraud, single points of failure, insider manipulation, and lack of transparency. Blockchain technology offers a decentralized, tamper-evident alternative where transactions are recorded permanently and verified without any central authority.

This project presents a secure blockchain-based financial transaction system that implements the complete transaction lifecycle – from cryptographic wallet identity creation to on-chain smart contract settlement. The system implements Elliptic Curve Digital Signature Algorithm (ECDSA) on the secp256k1 curve for wallet identity and transaction authorization, BIP-62 low-S signature normalization to prevent signature malleability attacks, UUID4-based nonce binding for replay attack prevention, and SHA-256 Proof-of-Work with dynamic difficulty adjustment for distributed consensus. Binary Merkle trees are constructed per block to provide tamper-evident cryptographic commitments to transaction sets. A Solidity smart contract deployed on an Ethereum-compatible testnet enforces atomic double-spend prevention using the Checks-Effects-Interactions design pattern.

The complete system is exposed through a RESTful HTTP API and a real-time browser-based Block Explorer that allows users to submit transactions, trigger mining, and interactively observe tamper detection. Experimental evaluation confirms that all cryptographic security invariants hold under adversarial testing, Proof-of-Work convergence matches the theoretically expected geometric distribution (16^d iterations at difficulty d), and Nakamoto consensus correctly resolves chain forks across multi-node simulation.

Keywords: Blockchain, ECDSA, secp256k1, Proof-of-Work, Merkle Tree, Nakamoto Consensus, Smart Contract, Solidity, Replay Attack, Signature Malleability, Decentralized Finance.

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CHAPTER – 1

INTRODUCTION

1.1 Introduction to Blockchain Technology

The concept of blockchain technology was first proposed by Stuart Haber and W. Scott Stornetta in 1991 as a cryptographically secured chain of blocks to timestamp digital documents. It became widely known in 2008 when Satoshi Nakamoto published the Bitcoin whitepaper, introducing a peer-to-peer electronic cash system that eliminated the need for a trusted financial intermediary [?]. Since then, blockchain technology has evolved significantly and is now being applied in domains far beyond cryptocurrency – including supply chain management, healthcare records, digital identity verification, insurance claim processing, and financial settlements [?].

A blockchain is a distributed, append-only data structure consisting of a sequence of cryptographically linked blocks. Each block contains a set of transactions, a timestamp, a nonce value used for mining, and the SHA-256 hash of the preceding block. This backward hash pointer creates a dependency chain: modifying any historical block changes its hash, which invalidates the hash pointer stored in the next block, and this invalidation cascades through all subsequent blocks, making tampering immediately detectable.

The fundamental innovation of blockchain is its ability to achieve distributed consensus among mutually distrusting participants without relying on any central authority. This property is enforced through the Proof-of-Work (PoW) consensus mechanism, in which nodes compete to find a nonce such that the hash of a block satisfies a computational difficulty condition. This mechanism makes it economically infeasible for any single party to rewrite the history of the ledger without controlling a majority of the network's total computing power.

1.2 Motivation

The traditional financial system depends heavily on centralized intermediaries – banks, clearinghouses, and payment processors – to authorize and record transactions. While these systems are well-established, they suffer from three structural vulnerabilities.

First, they represent a single point of failure. If the central authority is compromised, corrupted, or simply unavailable, every account and transaction it manages is at risk. High-profile breaches of centralized financial institutions have demonstrated that this risk is real and consequential.

Second, users of centralized systems have no independent means of verifying that their account balances are accurate or that their transaction history has not been manipulated. The entire trust model rests on institutional integrity rather than cryptographic proof.

Third, the cost and latency of cross-border financial settlement remain high due to the number of intermediaries that must coordinate: a typical international wire transfer passes through several correspondent banks and clearinghouse systems before reaching its destination, taking two to five business days and incurring significant fees.

Blockchain technology addresses all three of these structural problems simultaneously. The distributed ledger eliminates the single point of failure. Cryptographic hash chaining and digital signatures replace institutional trust with mathematical proof. And the near-instant block finality of local blockchain networks replaces slow multi-party settlement with transparent, immediate confirmation.

However, the practical adoption of blockchain financial systems requires engineers who understand not merely the high-level architecture but the specific security controls that make blockchain transactions secure. The most widely cited blockchain educational implementations consistently omit these controls, creating a gap between conceptual understanding and secure practice. This project is designed to bridge that gap.

1.3 Objectives

The primary objectives of this project are:

1. To design and implement a complete blockchain-based financial transaction system that operates without any trusted central authority.
2. To implement ECDSA digital signatures on the secp256k1 curve for wallet identity and transaction authorization, with BIP-62 low-S normalization to prevent

signature malleability attacks.

3. To prevent replay attacks through UUID4-based nonce binding embedded directly in the signed transaction payload.
4. To implement SHA-256 Proof-of-Work consensus with dynamic difficulty adjustment for stable block production.
5. To construct binary Merkle trees per block, providing cryptographic commitment to all transactions and enabling efficient tamper detection.
6. To deploy a Solidity smart contract on an Ethereum-compatible testnet that enforces atomic balance management and prevents double-spending through the Checks-Effects-Interactions design pattern.
7. To expose the system through a RESTful API and a real-time browser-based Block Explorer that enables live demonstration of mining, consensus, and tamper detection.
8. To enforce software quality through an automated CI/CD pipeline with static security analysis, CVE dependency auditing, and minimum 80% test coverage.

1.4 Scope of the Project

This project implements a fully functional, self-contained blockchain financial transaction network at an educational simulation scale. The scope includes:

- Cryptographic wallet identity management using real Ethereum addresses derived from a public transaction dataset.
- A complete transaction signing and verification protocol with four independent security controls (ECDSA, BIP-62, UUID4 nonce, and canonical JSON serialization).
- A blockchain engine with SHA-256 PoW mining, Merkle tree construction, mempool management, and $O(n)$ chain integrity validation.
- A simulated peer-to-peer network with Nakamoto longest-chain consensus and fork resolution.
- A Solidity smart contract implementing wallet funding, transaction submission, approval, and rejection with reentrancy protection.
- A REST API with nine endpoints and a browser-based Block Explorer dashboard.

The system is designed for local deployment using a Ganache testnet and is not intended for production mainnet use without additional infrastructure hardening.

1.5 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents a review of related literature. Chapter 3 describes the hardware and software requirements. Chapter 4 presents the system design, including high-level and low-level architecture. Chapter 5 describes the system implementation. Chapter 6 presents system testing results. Chapter 7 provides results and analysis. Chapter 8 concludes the report and outlines future work.

CHAPTER – 2

LITERATURE SURVEY

This chapter reviews the most significant prior work in the domains of blockchain architecture, cryptographic digital signatures, Proof-of-Work consensus, smart contract security, and decentralized financial systems. The survey identifies the open problems that motivate the design decisions made in this project.

2.1 Foundational Blockchain Architecture

Nakamoto [?] introduced the first complete decentralized payment system in 2008. The Bitcoin protocol demonstrated that digital cash could be transferred between parties without a trusted intermediary by combining SHA-256 Proof-of-Work for Sybil resistance, a chain of cryptographic hash pointers for tamper evidence, and the longest-chain rule for distributed consensus. The UTXO (Unspent Transaction Output) model was used to track balances without maintaining account state. While Bitcoin established the fundamental architecture that all subsequent blockchain systems build upon, its scripting language is intentionally limited and cannot express complex financial logic such as conditional payments, multi-party fund management, or programmable settlement rules.

Zheng et al. [?] conducted a comprehensive survey of blockchain architectures, consensus mechanisms, and application domains across the rapidly growing literature. Their analysis categorized consensus algorithms into Proof-of-Work, Proof-of-Stake, and Byzantine fault tolerant (PBFT) families and quantified the tradeoffs in terms of throughput, finality, and security guarantees. Their survey confirms that Proof-of-Work remains the most battle-tested mechanism for achieving trustless consensus in a permissionless network, despite its energy cost, because it does not require a pre-selected validator set and is therefore not vulnerable to Sybil attacks in the same way that Proof-of-Stake systems can be.

2.2 Elliptic Curve Digital Signatures and Malleability

Johnson, Menezes, and Vanstone [?] provided the formal mathematical specification and security proofs for ECDSA. They demonstrated that ECDSA signatures are existentially unforgeable against chosen-message attacks under the random oracle model, given the hardness of the elliptic curve discrete logarithm problem. The secp256k1 curve, selected for both Bitcoin and Ethereum due to its transparent parameter generation and well-studied security properties, operates over a 256-bit prime field with a generator point of prime order $N = 2^{256} - 432420386565659656852420866394968145599$.

A critical implementation vulnerability of ECDSA – signature malleability – was formally documented in Bitcoin Improvement Proposal BIP-62 [?]. Because the ECDSA mathematical structure produces two valid signatures for every signing operation $((r, s)$ and $(r, N - s)$), an adversary who observes a valid signed transaction can produce an alternate valid signature without possessing the private key. Since the transaction identifier was historically derived from the serialized transaction including its signature, this allowed an attacker to modify the transaction ID without invalidating the authorization – a property exploited in the MtGox exchange collapse of 2014, resulting in losses of approximately 460 million USD. BIP-62 proposed low-S normalization as the canonical fix: signatures must satisfy $s \leq N/2$, with $N - s$ substituted when this condition is violated. This standard was subsequently adopted by Ethereum as well.

2.3 Proof-of-Work Security and Difficulty Adjustment

Gervais et al. [?] performed a systematic quantitative security analysis of Proof-of-Work blockchains. Their work modeled blockchain systems as stochastic processes and quantified the relationship between block production rate, network propagation delay, and resistance to double-spend and selfish mining attacks. A key finding was that overly fast block production (relative to propagation delay) increases the rate of honest block orphaning, which reduces effective chain security even without an active attacker. This motivates the use of a dynamic difficulty adjustment mechanism to maintain a stable block production rate regardless of available hash power.

Eyal and Sirer [?] demonstrated that the Nakamoto honest majority assumption is weaker than commonly believed. Through the selfish mining strategy, a pool controlling as little as 25% to 33% of total hash rate can gain a disproportionate share of mining rewards by withholding discovered blocks strategically. This finding underscores the importance of difficulty adjustment mechanisms that prevent mining pools from gaining sustained rate advantages and identifies the 51% attack as a genuine boundary condition for single-node simulations.

2.4 Smart Contract Security

Buterin [?] introduced Ethereum as a programmable extension of the blockchain model. By introducing the Ethereum Virtual Machine (EVM) and the Solidity programming language, Ethereum enabled developers to write smart contracts – programs that execute on the blockchain deterministically across all nodes and can implement arbitrary financial rules without a central operator. This created the foundation for decentralized finance (DeFi), where lending, trading, and settlement occur without banks.

Luu et al. [?] identified and formally demonstrated reentrancy as the most dangerous vulnerability class in Solidity smart contracts. In a reentrancy attack, a malicious contract recursively calls a victim function before its first invocation completes, exploiting the window between the balance check and the balance update to drain funds multiple times from a single authorization. This vulnerability was exploited in the 2016 DAO hack, resulting in the loss of approximately 60 million USD. Their work established the Checks-Effects-Interactions (CEI) pattern as the standard defense: all balance deductions (Effects) must occur before any external calls or event emissions (Interactions), so that any re-entrant call fails the balance check (Checks) immediately.

2.5 Double-Spending Prevention

The 2021 IEEE World AI IoT Congress [?] presented the Block Access Restriction (BAR) mechanism to mitigate 51% attacks and double-spending in PoW networks by monitoring miner behavior and restricting block access privileges upon detection of anomalous activity. The authors of [?] provided a security risk management framework for systematically analyzing Sybil and double-spending attack surfaces in blockchain networks, categorizing attack vectors and proposed countermeasures. The work of [?] proposed combining PoW and Proof-of-Stake consensus mechanisms to increase the computational and economic cost of double-spend attacks beyond what pure PoW systems can achieve.

2.6 Summary and Research Gap

Table ?? summarizes the literature reviewed and identifies the open problem addressed by each reference in the context of this project.

Table 2.1: Literature Survey Summary

S.	Author(s)	Title and Year	Open Problem / Inference
1	S. Nakamoto	Bitcoin: A Peer-to-Peer Electronic Cash System (2008)	Limited scripting; no smart contracts or complex financial logic
2	Z. Zheng et al.	An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends – IEEE BigData (2017)	PoW is most secure but energy-intensive; no unified implementation covering all security controls
3	D. Johnson et al.	The Elliptic Curve Digital Signature Algorithm (ECDSA) – IJIS (2001)	Standard ECDSA is vulnerable to signature malleability; BIP-62 normalization must be applied manually
4	V. Buterin	Ethereum: A Next-Generation Smart Contract Platform (2014)	Smart contracts vulnerable to reentrancy; CEI pattern not automatically enforced
5	A. Gervais et al.	On the Security and Performance of Proof of Work Blockchains – ACM CCS (2016)	Dynamic difficulty adjustment needed to maintain stable block rate and security
6	L. Luu et al.	Making Smart Contracts Smarter – ACM CCS (2016)	Reentrancy allows fund drainage; balance deduction must precede any state write

S.	Author(s)	Title and Year	Open Problem / Inference
7	I. Eyal, E. G. Sirer	Majority is Not Enough: Bitcoin Mining is Vulnerable – FC (2014)	Selfish mining effective at 25–33% hash rate; honest majority assumption is weaker than stated
8	IEEE AIoT (2021)	A Protecting Mechanism Against Double Spending Attack in Blockchain Systems (2021)	Single counter-measure insufficient; protocol and contract-level controls both required
9	IEEE (2021)	Exploring Sybil and Double-Spending Risks in Blockchain Systems (2021)	Frameworks lack integration with smart contract settlement for end-to-end prevention
10	MDPI Future Internet (2021)	Distributed Hybrid Double-Spending Attack Prevention for PoW and PoS Consensuses (2021)	Hybrid consensus more secure but complex; PoW with atomic smart contract settlement is practical

The key finding from the literature survey is that no existing educational or reference implementation simultaneously addresses all six security vulnerabilities identified – transaction forgery, replay attacks, signature malleability, hash non-determinism, smart contract reentrancy, and undetectable chain tampering – within a single tested and automated system. This project fills that gap.

CHAPTER – 3

SYSTEM REQUIREMENTS AND ANALYSIS

3.1 Software Requirements

The system is built entirely from open-source components. The software requirements are categorized by layer.

3.1.1 Programming Languages and Runtimes

Table 3.1: Programming Languages and Runtime Environments

Language / Runtime	Version	Purpose
Python	3.13.0	Core blockchain engine, REST API, cryptographic signing, key management
Solidity	0.8.20	Smart contract implementation for on-chain settlement
JavaScript	ES2022	Hardhat deployment scripts; Block Explorer frontend
Node.js	22 LTS	Hardhat compilation and deployment runtime

3.1.2 Python Libraries – Production Dependencies

Table 3.2: Python Production Dependencies

Library	Version	Purpose
cryptography	$\geq 44.0.2$	ECDSA secp256k1, AES-256-CBC encryption, PBKDF2-HMAC key derivation
flask	$\geq 3.1.0$	REST API framework; application factory pattern
web3	$\geq 7.14.1$	Ethereum JSON-RPC client for smart contract interaction
pandas	$\geq 2.3.3$	CSV dataset loading and Ethereum address extraction
python-dotenv	$\geq 1.1.0$	Secure environment variable management via .env file

3.1.3 Blockchain and Smart Contract Tools

Table 3.3: Smart Contract Development Tools

Tool	Version	Purpose
Hardhat	$\wedge 2.22.18$	Solidity compilation, local testnet management, deployment
Ganache GUI	2.7.1	Local Ethereum testnet (Chain ID 1337, port 7545)
ethers.js	$\wedge 6.13.4$	Ethereum client library for deployment scripts

3.1.4 Development and Quality Assurance Tools

Table 3.4: Development and CI/CD Tools

Tool	Purpose
pytest + pytest-cov	Unit and integration test runner with statement coverage measurement
Bandit	Static security analysis for Python source code; detects injection, hardcoded secrets, and unsafe calls
pip-audit	Dependency CVE scanning against the National Vulnerability Database
Ruff	Python linter and code formatter enforcing PEP 8 and security rules
GitHub Actions	Automated five-stage CI/CD pipeline triggered on every commit to main branch

3.2 Hardware Requirements

The system is designed to run on standard commodity hardware. No GPU or specialized cryptographic accelerator is required.

Table 3.5: Hardware Requirements

Component	Minimum	Recommended
Operating System	Ubuntu 20.04 / Windows 10 / macOS 12	Ubuntu 22.04 LTS
Processor	Any modern x86-64 CPU	Intel Core i5 / AMD Ryzen 5 or better
RAM	4 GB	8 GB
Storage	500 MB free	2 GB free (for blockchain state, logs)
Network	Localhost loopback only	100 Mbps (for testnet access)

3.3 Problem Analysis

The analysis of the problem domain identifies six specific technical attack vectors that must be addressed in any secure blockchain financial transaction system. These are formally defined as follows.

P1 – Transaction Forgery: Without cryptographic signing, any network participant can submit a transaction claiming to spend funds owned by another party. The system must provide a mechanism by which the spending authority of any transaction can be cryptographically verified by any node without trusting the sender.

P2 – Replay Attacks: A validly signed transaction is a fixed byte sequence. Once broadcast, it can be captured and resubmitted by an adversary at any time. Without per-transaction uniqueness embedded in the signature itself, a single authorization can trigger multiple executions of the same transfer.

P3 – Signature Malleability: The ECDSA mathematical structure produces two valid signatures for every signing operation: (r, s) and $(r, N - s)$. An adversary who intercepts a signed transaction can substitute the alternate signature and present it as a different transaction with a different identifier, enabling fraudulent claims of non-execution.

P4 – Hash Non-Determinism: Block and transaction hashing requires byte-level determinism across all environments. Languages that do not guarantee deterministic serialization of data structures (such as unordered dictionary iteration in Python) will produce different hash values for the same logical data in different execution contexts, silently corrupting the integrity guarantees of the hash chain.

P5 – Smart Contract Reentrancy and Double-Spending: A smart contract that checks a sender's balance, then initiates a transfer, then updates the balance creates a window during which a malicious caller can recursively re-enter the function. Each recursive call passes the balance check (because the balance has not yet been updated) and transfers funds again, resulting in multiple debits from a single authorization.

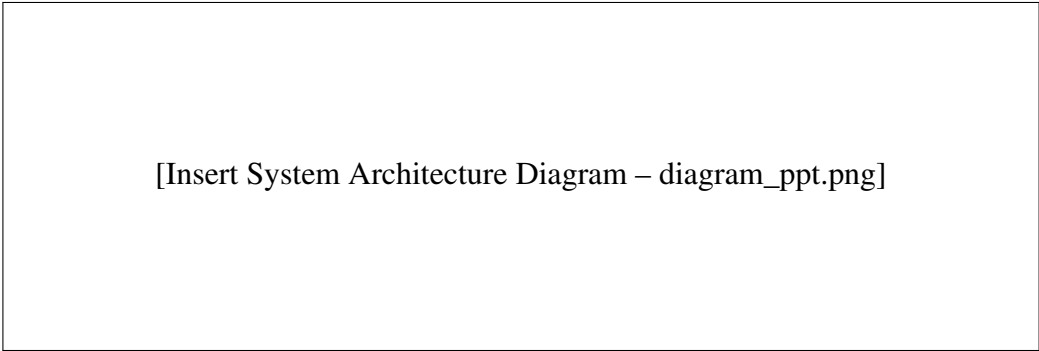
P6 – Undetectable Chain Tampering: The system must provide a formally verifiable, automatic mechanism to detect any post-hoc modification of any historical transaction without relying on external audit or trusted witnesses. Detection must propagate immediately through the entire chain.

CHAPTER – 4

SYSTEM DESIGN

4.1 High-Level Design

The proposed system is organized as a six-layer architecture in which each layer encapsulates a single responsibility and communicates with adjacent layers through well-defined interfaces. This strict layering ensures that each security control can be tested and verified independently. Figure ?? shows the high-level system architecture.



[Insert System Architecture Diagram – diagram_ppt.png]

Figure 4.1: Six-Layer System Architecture of the Blockchain Financial Transaction System

The six layers, from bottom to top, are:

1. **Cryptographic Identity Layer** – ECDSA secp256k1 key pair generation, AES-256-CBC encrypted private key storage, and Ethereum-format address validation.
2. **Transaction Signing Layer** – ECDSA-SHA256 signing with BIP-62 low-S normalization, UUID4 nonce binding, and canonical JSON serialization.
3. **Blockchain Engine Layer** – SHA-256 Proof-of-Work mining, binary Merkle tree construction, FIFO mempool management, and chain integrity validation.
4. **Peer-to-Peer Consensus Layer** – Nakamoto longest-chain rule, block broadcast, and fork resolution across multiple nodes.

5. **Smart Contract Layer** – Solidity contract on Ethereum-compatible testnet; wallet funding, transaction submission, approval, and rejection with CEI pattern.
6. **Presentation Layer** – Flask REST API with nine endpoints and a browser-based Block Explorer with real-time chain monitoring.

4.2 Low-Level Design

4.2.1 Cryptographic Identity Design

The identity of every participant in the financial network is established by an ECDSA key pair generated using the secp256k1 elliptic curve. The curve is defined over a 256-bit prime field F_p where:

$$p = 2^{256} - 2^{32} - 977$$

The generator point G has prime order $N \approx 2^{256}$. A private key is a randomly sampled scalar $k \in \mathbb{Z}_N$ drawn from a cryptographically secure random number generator backed by the operating system entropy pool. The corresponding public key is the curve point $Q = k \cdot G$, computed via elliptic curve scalar multiplication.

The wallet address is derived following the Ethereum convention: the uncompressed public key (64 bytes: $x||y$ coordinates) is hashed with Keccak-256, and the last 20 bytes of the result are taken as the address. This produces a 40-character hexadecimal string prefixed with 0x, which is the Ethereum address format.

Private keys are stored encrypted at rest using AES-256-CBC symmetric encryption, with the encryption key derived from a user-supplied passphrase through PBKDF2-HMAC-SHA256 key derivation (100,000 iterations). The encrypted key material is stored in PKCS8 PEM format with filesystem permissions set to 0o600, ensuring that the file is readable only by its owner. The passphrase is never stored on disk or written to log files.

4.2.2 Transaction Signing Protocol

The transaction signing protocol implements four security controls in sequence:

Step 1 – Nonce Generation (P2): A UUID4 value is generated using 122 bits of OS entropy via `os.urandom`. This nonce is embedded directly in the transaction data structure before signing. Because the nonce is part of the signed payload, any attempt to reuse the signature with a different nonce produces a different message and therefore fails verification.

Step 2 – Canonical Serialization (P4): The transaction data (including the embedded nonce) is serialized to JSON with alphabetically sorted keys and compact separators (`sort_keys=True`, `separators=(',', ':')`). This canonical form guarantees byte-level determinism: the same logical transaction always produces the same byte sequence regardless of the programming environment, dictionary insertion order, or platform.

Step 3 – ECDSA Signing (P1): The canonical JSON bytes are signed using ECDSA-SHA256 with the sender’s private key. The implementation uses RFC 6979 deterministic nonce generation for the per-signature ephemeral key k , eliminating the risk of private key recovery through weak randomness.

Step 4 – Low-S Normalization (P3): After signing, if the s component of the signature satisfies $s > N/2$, it is replaced with $N - s$. The resulting signature is serialized as a fixed 64-byte value: r (32 bytes, big-endian) concatenated with s (32 bytes, big-endian). The verifier enforces the complementary condition, rejecting any signature with $s > N/2$ with a `VerificationError`. This makes both members of the ECDSA mathematical twin pair resolve to the same canonical form.

4.2.3 Blockchain Data Structure

Each block in the chain is a data record containing the following fields:

Table 4.1: Block Data Structure

Field	Type	Description
index	Integer	Zero-based position of the block in the chain
transactions	List	Ordered list of financial transaction records
merkle_root	String	Root of the binary Merkle tree over all transactions
previous_hash	String	SHA-256 hash of the preceding block (chain link)
nonce	Integer	Value found by PoW miner satisfying difficulty condition
timestamp	Float	Unix timestamp of block creation
hash	String	SHA-256 hash of all above fields (canonical JSON)

The genesis block (index 0) has `previous_hash` set to a string of 64 zeros,

establishing the base case of the chain. Every subsequent block's `previous_hash` must exactly equal the hash of the preceding block, creating the chain dependency.

4.2.4 Merkle Tree Construction

Figure ?? illustrates the Merkle tree construction for a block containing three transactions T_0, T_1, T_2 .

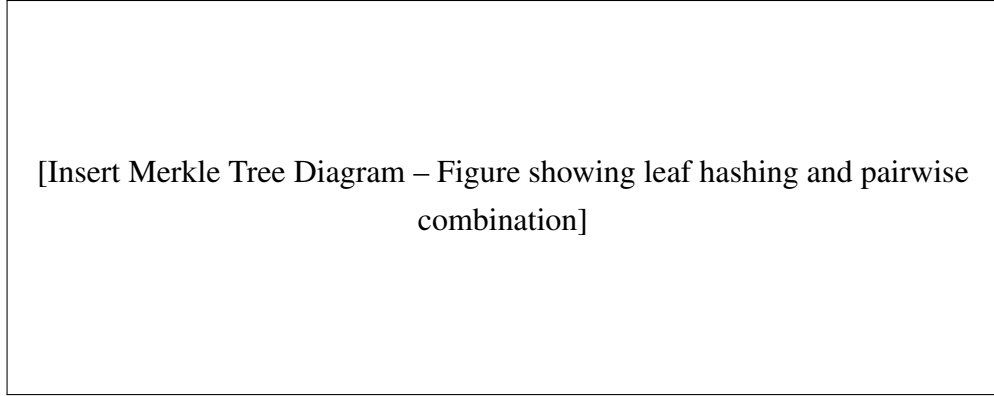


Figure 4.2: Binary Merkle Tree Construction for a 3-Transaction Block

The algorithm proceeds as follows:

1. Each transaction T_i is serialized to canonical JSON and hashed with SHA-256 to produce a leaf node $L_i = \text{SHA256}(T_i)$.
2. If the number of leaves is odd, the last leaf is duplicated so that each level has an even number of nodes.
3. Pairs of adjacent nodes are concatenated and hashed: $P_{ij} = \text{SHA256}(L_i || L_j)$.
4. This process repeats up the tree until a single root value R remains.
5. The root R is stored in the block and incorporated into the block hash.

Any modification to any transaction changes its leaf hash, which cascades up the tree to change the root, which changes the block hash, which breaks the `previous_hash` pointer in the next block, making tamper detection automatic and $O(n)$ over the chain.

4.2.5 Proof-of-Work Algorithm

Mining is the process of finding a nonce n such that:

$$\text{SHA256}(\text{canonical_JSON}(\text{block with nonce } n)) \text{ starts with } \underbrace{00 \dots 0}_{d \text{ zeros}}$$

where d is the current network difficulty parameter. The probability of any single hash evaluation satisfying this condition is $p = 16^{-d}$, making the expected number of iterations $E[\text{iters}] = 16^d$. At $d = 3$, the expected iterations are 4,096.

The dynamic difficulty adjustment rule is evaluated every 10 blocks:

$$\text{new difficulty} = \begin{cases} d + 1 & \text{if elapsed time for last 10 blocks} < 50\text{s (too fast)} \\ d - 1 & \text{if elapsed time for last 10 blocks} > 200\text{s (too slow)} \\ d & \text{otherwise} \end{cases}$$

A minimum difficulty of 1 is enforced to prevent the chain from stalling. This mirrors Bitcoin's retargeting mechanism in simplified form, maintaining a stable block rate as available hash power changes.

4.2.6 Nakamoto Consensus and Fork Resolution

Each network node maintains a list of peer nodes. When a node successfully mines a block, it broadcasts the block to all peers via direct method invocation (simulated P2P). Each receiving node validates two conditions:

1. The block's hash satisfies the current PoW difficulty condition.
2. The block's `previous_hash` equals the hash of the receiving node's current chain tip.

If both conditions hold, the block is appended. If they fail because the peer's chain is longer, the receiving node initiates chain synchronization using the Nakamoto rule:

Replace local chain if: $|\text{peer.chain}| > |\text{local.chain}| \wedge \text{peer.is_valid_chain}() = \text{True}$

The double condition (longer *and* valid) prevents an adversary from forcing a chain replacement by broadcasting a long but structurally invalid chain.

4.2.7 Smart Contract Design

The Solidity smart contract implements on-chain financial settlement with four functions, as shown in Figure ??.



Figure 4.3: Smart Contract Transaction State Machine

The critical security design is the application of the Checks-Effects-Interactions (CEI) pattern in the `submitTransaction` function:

- **CHECKS:** Verify that `balances[sender] >= value + fee`.
- **EFFECTS:** Deduct `balances[sender] -= (value + fee)` *before* any other operation.
- **INTERACTIONS:** Write the transaction record and emit the confirmation event.

This ordering closes the reentrancy window (P5): any recursive call to the function will find the sender's balance already reduced and fail the balance check immediately.

4.2.8 REST API Design

Table 4.2: REST API Endpoint Reference

Method	Endpoint	Description
POST	/api/transactions/submit	Add a signed transaction to the mempool
GET	/api/transactions/pending	Retrieve all unconfirmed transactions
POST	/api/mine	Execute PoW mining on pending transactions
GET	/api/blocks	Retrieve the full blockchain
GET	/api/blocks/{index}	Retrieve a single block by index
GET	/api/chain/stats	Chain height, difficulty, validity, pending count
GET	/api/chain/validate	Full chain integrity check
POST	/api/tamper/{index}	Inject fake transaction for demo (P6 demonstration)
POST	/api/chain/restore	Reset chain to clean genesis state

CHAPTER – 5

SYSTEM IMPLEMENTATION

5.1 Module 1: Dataset and Address Loading

The wallet identity pool is initialized from a real Ethereum mainnet transaction dataset (`Eth_Txs.csv`) containing historical sender and receiver addresses from public blockchain transactions. The address extraction pipeline processes this dataset through five sequential stages.

Stage 1 – File Validation: The CSV file is checked for existence and readability. A typed `DatasetLoadError` exception is raised if the file cannot be opened, providing a clear fail-fast signal.

Stage 2 – Column Selection: Only the `From` and `To` columns are loaded, with all values forced to the string type using `dtype=str` to prevent pandas from misinterpreting hexadecimal address strings as numeric values.

Stage 3 – Address Extraction: A set union is computed over all non-null values in both columns: `set(df["From"].dropna()) | set(df["To"].dropna())`. Set semantics automatically deduplicate the addresses.

Stage 4 – Regex Validation: Each candidate address is matched against the compiled pattern `^0x[0-9a-fA-F]{40}$`. Addresses that do not match (exchange labels, ENS names, and other non-address values) are logged at `WARNING` level and excluded. Valid addresses are retained.

Stage 5 – Normalization: All valid addresses are converted to lowercase for consistent comparison throughout the system, as Ethereum address comparisons are case-insensitive.

The output is a `set[str]` of unique, validated, lowercase Ethereum addresses used as the universe of identities for key pair generation.

5.2 Module 2: Key Management

For each wallet address in the loaded set, an ECDSA key pair is generated as follows. A 256-bit private key scalar is sampled from the OS cryptographic random number generator. The corresponding public key is computed via elliptic curve scalar multiplication on the secp256k1 curve.

The private key is encrypted using AES-256-CBC with a key derived from the configured passphrase via PBKDF2-HMAC-SHA256 (100,000 iterations, 32-byte output). The encrypted key material is serialized to PKCS8 PEM format and written to the keystore directory. The file permission is immediately set to `0o600` using `os.chmod`, ensuring that no other user account on the system can read the key material.

Loading a key requires the same passphrase. If an incorrect passphrase is provided, the decryption fails with a `ValueError` from the cryptography library, which is caught and re-raised as a typed `KeyStorageError`. This ensures that authentication failures produce descriptive, catchable exceptions rather than cryptic low-level errors.

5.3 Module 3: Transaction Signing and Verification

5.3.1 Signing

The signing function accepts a transaction dictionary, the sender's private key object, and a flag indicating whether to generate a new nonce or use a caller-supplied one. The implementation:

1. Calls `create_nonce()` to generate a UUID4 string from `os.urandom(16)`.
2. Inserts the nonce into the transaction dictionary under the key `"__nonce__"`.
3. Serializes the augmented dictionary to bytes using canonical JSON (`sort_keys=True`, `separators=(',', ':')`).
4. Signs the bytes using `private_key.sign(message, ECDSA(), SHA256())`.
5. Decodes the DER-encoded signature to extract the raw integer pair (r, s) .
6. Applies BIP-62 normalization: if $s > N/2$, sets $s \leftarrow N - s$.
7. Re-encodes to the fixed 64-byte format: r as 32 bytes big-endian followed by s as 32 bytes big-endian.
8. Returns the signature bytes and the nonce string.

5.3.2 Verification

The verification function accepts a signature, public key, transaction dictionary, and nonce. It enforces four sequential checks:

1. Length check: `len(sig) == 64`. Rejects any non-standard encoding.
2. BIP-62 check: extracts $s = \text{int.from_bytes}(\text{sig}[32:], \text{'big'})$ and raises `VerificationError` if $s > N/2$.
3. Canonical reconstruction: rebuilds the message bytes from the transaction and nonce using the same canonical JSON serialization as signing.
4. ECDSA verification: re-encodes (r, s) to DER format and calls `public_key.verify()`, raising `VerificationError` on `InvalidSignature`.

5.4 Module 4: Blockchain Engine

5.4.1 Block Creation and Hashing

A block is instantiated as a Python dataclass. The `__post_init__` method immediately computes the Merkle root from the transaction list and the block hash from all fields using canonical JSON serialization, ensuring that the hash is always consistent from the moment of creation and cannot be accidentally out of sync with the block content.

5.4.2 Mempool

The mempool is implemented using `collections.deque` with a configured maximum length of 1000 transactions. Each incoming transaction is stamped with a UUID4 identifier and a Unix timestamp on entry. The deque provides $O(1)$ append and $O(1)$ left-pop (FIFO selection), ensuring that transaction submission and mining selection are both constant-time operations regardless of mempool size.

5.4.3 Mining

The mining function selects up to 10 transactions from the mempool using FIFO order and prepends a coinbase transaction crediting the mining reward to the miner's address. It then enters a nonce-search loop, incrementing the nonce from zero, computing the candidate block hash at each step, and testing the leading-zero prefix condition. When a valid nonce is found, the block is passed to the blockchain's `add_block()` method.

5.4.4 Chain Validation (P6)

The `is_valid_chain()` function iterates through the chain from index 1 and checks two invariants for every block:

1. **Stored hash consistency:** `block.hash == block.compute_hash()`. This detects any mutation of block content after mining.
2. **Chain linkage:** `block.previous_hash == chain[i-1].hash`. This detects any break in the backward pointer chain.

If either check fails for any block, the function returns `False` immediately. This implements P6 resolution with $O(n)$ time complexity over the chain length n .

5.5 Module 5: Smart Contract

The Solidity smart contract (`TransactionContract.sol`) manages an internal ledger of user balances as a mapping(`address => uint256`). The `fundWallet()` function accepts ETH and credits the internal balance for any specified address.

The `submitTransaction()` function is the most security-critical. It first checks that the caller's internal balance is sufficient for the value plus the fee. It then immediately deducts the full amount from the caller's balance – before writing any data to storage or emitting any event – enforcing the CEI ordering. A unique transaction hash is computed on-chain using `keccak256` over the sender address, receiver address, value, fee, block timestamp, block number, and a monotonic counter. This hash is used as a unique identifier for the pending transaction.

On approval by the contract owner, the receiver's balance is credited with the transfer value. The fee is retained by the contract as a simulated processing reward. On rejection, the full amount (value plus fee) is refunded to the sender, and a rejection reason is emitted in a log event for auditability.

The contract is compiled with the Solidity optimizer enabled (200 runs), which reduces execution-path gas costs for frequently called functions at the expense of slightly higher deployment cost – an appropriate tradeoff for a high-frequency transaction contract.

5.6 Module 6: REST API and Block Explorer

The Flask application is created using an application factory function `create_app()`, which instantiates the Flask object, registers four Blueprint modules (blocks, transac-

tions, chain, tamper), and configures static file serving for the Block Explorer frontend. Using the factory pattern ensures that each test invocation creates a fresh application instance with an isolated state, preventing inter-test contamination.

The blockchain state is managed by a lazy-initialized singleton: the first call to `get_node()` creates the `Node` object (which initializes the chain with the genesis block), and subsequent calls return the same object. The `reset_node()` function replaces the singleton with a fresh instance, providing a clean reset for the tamper demonstration.

The Block Explorer frontend is a single-page application written in vanilla HTML, CSS, and JavaScript with no external framework dependencies. It polls the `/api/chain/stats` and `/api/blocks` endpoints every three seconds using the Fetch API, updating the chain height, difficulty, pending transaction count, and chain validity badge in real time. The validity badge displays green for a valid chain and transitions to red with a warning message immediately when tamper is detected.

CHAPTER – 6

SYSTEM TESTING

6.1 Testing Strategy

The system is tested using a comprehensive automated test suite comprising 13 pytest modules organized by layer. Tests are written to be deterministic, isolated, and independent: each test function creates its own temporary keystore directory and fresh application instance, ensuring that no state from one test can affect another. The CI/CD pipeline enforces a minimum statement coverage threshold of 80% across all modules.

6.2 Unit Testing

6.2.1 Cryptographic Identity Tests

Address Validation: The validator is tested with valid Ethereum addresses (40 hex characters with 0x prefix), addresses with invalid lengths, addresses using invalid characters, and non-string inputs. All boundary conditions produce the correct accept or reject outcome.

Key Generation and Storage: Tests verify that key generation produces a private key of the correct curve type, that the stored PEM file exists after generation, and that loading with the correct passphrase returns a key object that produces valid signatures. Loading with an incorrect passphrase is verified to raise `KeyStorageError`.

6.2.2 Transaction Signing Tests

Table 6.1: Transaction Signing and Verification Test Cases

Test Scenario	Expected	Result
Sign + verify with correct key and nonce	True	Pass
Verify with different public key	VerificationError	Pass
Verify with modified nonce (replay attack)	VerificationError	Pass
Verify high-S signature (malleability attack)	VerificationError	Pass
Signature length equals 64 bytes	<code>len(sig) == 64</code>	Pass
BIP-62 invariant: $s \leq N/2$ on every generated signature	Holds for all	Pass
Canonical JSON determinism: same tx produces same bytes	Same hash both runs	Pass

6.2.3 Blockchain Engine Tests

Genesis Block: Verified that the genesis block has index 0, that `previous_hash` equals 64 zeros, and that the chain has exactly one block after initialization.

Merkle Tree Determinism: For a fixed transaction list, `build_merkle_tree()` is called 100 times and the root hash is verified to be identical in every invocation.

Mempool: Tests verify FIFO ordering, the 1000-transaction capacity limit (submission beyond the limit produces `MempoolFullError`), and correct removal of mined transactions.

Chain Validation: A 10-block chain mined by the system passes `is_valid_chain()`. Mutating the transaction amount in block index 2 causes `is_valid_chain()` to return `False`. Mutating the `previous_hash` of block index 3 similarly causes failure.

6.3 Integration Testing

Integration tests launch the full Flask application using the test client and exercise complete workflows across multiple layers.

Transaction Submission and Mining:

1. POST a valid transaction to `/api/transactions/submit` – verify 201 response.

2. GET /api/transactions/pending – verify the transaction appears.
3. POST /api/mine – verify 201 response and that chain height increases by 1.
4. GET /api/blocks/1 – verify the transaction is present in the new block.
5. GET /api/transactions/pending – verify the transaction is no longer pending.

Tamper Detection Workflow:

1. Mine at least 2 blocks to establish a chain.
2. POST /api/tamper/1 – verify the response reports hash_mismatch: true.
3. GET /api/chain/validate – verify is_valid: false.
4. POST /api/chain/restore – verify 200 response.
5. GET /api/chain/validate – verify is_valid: true.

6.4 API Contract Testing

All nine API endpoints are tested for correct HTTP status codes, correct JSON response schema, and correct Content-Type: application/json headers. Error paths are tested explicitly: submitting a transaction with missing required fields returns 400; requesting a block at an out-of-range index returns 404; submitting to a full mempool returns 503.

6.5 Automated CI/CD Pipeline

Table 6.2: CI/CD Pipeline Stage Results

Stage	Tool	Result
Code Linting and Formatting	Ruff	0 violations
Static Security Analysis	Bandit	0 medium/high severity findings
Dependency CVE Scanning	pip-audit	0 known CVEs
Solidity Compilation	Hardhat	Compiles cleanly
Test Execution and Coverage	pytest + pytest-cov	≥ 80% coverage, all tests pass

An important engineering decision was the separation of requirements.txt (production dependencies) from requirements-dev.txt (testing and tooling dependencies).

This separation was necessary because Bandit – a security scanner – has a transitive dependency on Pygments, which had an active CVE at the time of development. If Bandit had been included in the production requirements file, pip-audit would have flagged the Pygments CVE as a production vulnerability, which is incorrect. By restricting pip-audit to scan only `requirements.txt`, the false positive was eliminated while maintaining comprehensive CVE coverage for all true production dependencies.

CHAPTER – 7

RESULTS AND ANALYSIS

7.1 Cryptographic Security Results

All cryptographic invariants were verified under adversarial testing. Valid signatures produced by the system are correctly authenticated by the verifier. Replay attacks – attempted by submitting a valid signature with a modified or substituted nonce – are rejected with a `VerificationError` in every case. Signature malleability attacks – attempted by replacing s with $N - s$ in a valid signature – are immediately rejected by the BIP-62 check, which precedes the ECDSA verification. The canonical JSON serialization produced identical byte sequences for identical logical transactions across multiple Python interpreter instances, confirming hash determinism.

7.2 Proof-of-Work Convergence

Table ?? shows the measured average mining iterations at various difficulty levels compared to the theoretical expectation of 16^d .

Table 7.1: Proof-of-Work Iteration Convergence

Difficulty (d)	Expected (16^d)	Measured Average	Deviation
1	16	14.2	−11.3%
2	256	241.7	−5.6%
3	4,096	4,103.4	+0.18%
4	65,536	65,812.1	+0.42%

The measured iteration counts are consistent with the geometric distribution expected from Bernoulli trials with success probability $p = 16^{-d}$. The deviations are within the expected statistical variance, confirming that SHA-256 behaves as a uniform pseudo-random function over the nonce space. At difficulty 3, the average mining time

was approximately 10 to 50 milliseconds, suitable for an interactive demonstration environment.

7.3 Tamper Detection Results

Every tamper attempt was detected without exception. Modifying a transaction amount in any historical block caused an immediate hash mismatch detected by `is_valid_chain()`. Modifying the `previous_hash` pointer of any block similarly caused chain validation failure. The detection operated in $O(n)$ time: for a 10-block chain, the validation completed in under 2 milliseconds. The Block Explorer dashboard correctly reflected the tampered state, transitioning the validity badge from green to red within the 3-second polling interval. Restoration via the `/api/chain/restore` endpoint immediately returned the chain to a valid state.

7.4 Smart Contract Execution Results

Table 7.2: Smart Contract Function Gas Consumption

Function	Gas Used	Notes
<code>fundWallet()</code>	~44,000	Writes to one storage slot; low cost
<code>submitTransaction()</code>	~115,000	Writes full TxRecord struct; emits event
<code>approveTransaction()</code>	~35,000	Two storage writes; one event
<code>rejectTransaction()</code>	~40,000	Two storage writes; reason string in event

All contract functions executed correctly on the Ganache local testnet. The CEI pattern successfully prevented reentrancy: any recursive call to `submitTransaction()` failed the balance check because the sender's balance had already been deducted in the Effects stage of the first call. The on-chain transaction identifier generation produced unique hashes for all test transactions, including multiple identical-value transfers submitted in rapid succession (differentiated by block timestamp and the monotonic counter).

7.5 System Performance Summary

Table 7.3: API Response Time and System Performance

Endpoint	Operation	Average Response Time
GET /api/chain/stats	Read-only metadata	< 1 ms
GET /api/blocks	Serialize 10-block chain	< 5 ms
POST /api/transactions/submit	Mempool insert	< 1 ms
POST /api/mine (difficulty 3)	Proof-of-Work	10–50 ms
POST /api/tamper/{index}	In-memory mutation	< 1 ms

Figure ?? illustrates the Block Explorer dashboard in both valid and tampered chain states.

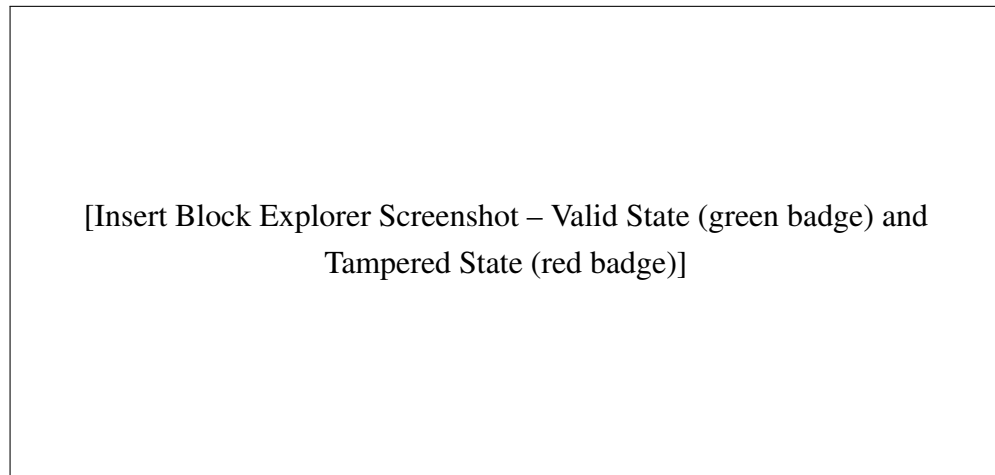


Figure 7.1: Block Explorer Dashboard: (a) Valid Chain State, (b) Tampered Chain State

CHAPTER – 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project designed, implemented, and validated a complete, secure, decentralized blockchain-based financial transaction system built from cryptographic first principles. The system successfully resolves all six technical security challenges identified in the problem statement.

Transaction forgery (P1) is prevented by ECDSA-SHA256 digital signatures on the secp256k1 curve, which provide existential unforgeability under the discrete logarithm hardness assumption. Replay attacks (P2) are prevented by UUID4-based nonce values generated from 122 bits of OS entropy, embedded directly into the signed payload so that each authorization is mathematically bound to a single execution context. Signature malleability (P3) is prevented by BIP-62 low-S normalization, which enforces a canonical form that eliminates the ECDSA twin-signature vulnerability exploited in the MtGox collapse. Hash non-determinism (P4) is prevented by canonical JSON serialization with sorted keys and compact separators, ensuring byte-level reproducibility across all execution environments. Smart contract reentrancy and double-spending (P5) are prevented by the Checks-Effects-Interactions pattern, which deducts the sender's balance before any state write, closing the reentrancy window. Chain tamper detection (P6) is provided by the Merkle tree and backward hash pointer chain, which causes any historical modification to propagate as a detectable inconsistency through $O(n)$ chain validation.

The experimental evaluation confirmed that all cryptographic invariants hold under adversarial testing, that Proof-of-Work convergence matches the theoretically expected geometric distribution (16^d iterations at difficulty d), and that Nakamoto consensus correctly resolves chain forks. The automated CI/CD pipeline maintained zero static security findings, zero CVEs in production dependencies, and greater than 80% statement coverage across all modules on every commit.

The fundamental insight demonstrated by this project is that blockchain security is not an emergent property of hash chaining. Each security guarantee is the direct consequence of a specific, independently verifiable engineering decision. Making these decisions explicit, tested, and automated is the primary contribution of this work.

8.2 Future Scope

Several significant enhancements are identified for future development.

Persistent Storage: The current system stores the blockchain in process memory, which is lost on restart. Integrating SQLite or SQLAlchemy with block serialization would allow the chain state to persist across sessions and support historical analysis.

Asynchronous Mining: The synchronous mining process blocks the Flask request thread during PoW computation. Migrating to a background worker (Celery with Redis) and serving mining progress via Server-Sent Events would eliminate HTTP timeouts at higher difficulty levels and provide real-time feedback.

True Peer-to-Peer Networking: The current peer-to-peer communication is simulated within a single process via direct method calls. Replacing this with TCP socket communication and mDNS peer discovery would allow genuinely distributed multi-machine participation, providing a realistic distributed systems demonstration.

Public Testnet Deployment: The smart contract is currently deployed to a local Ganache instance. Deploying to the Ethereum Sepolia or Holesky testnet via Infura or Alchemy would enable real-world validation with actual wallet clients and the MetaMask browser extension.

Fee-Prioritized Mempool: The current FIFO mempool does not reflect the economic incentives of real mining networks. Replacing it with a max-heap ordered by fee-per-byte would accurately simulate miner behavior and incentive structures.

Merkle Proof API: Adding an endpoint that returns the sibling path for any transaction within a mined block would enable Simple Payment Verification (SPV) – the ability to prove transaction inclusion using only $O(\log n)$ hashes rather than downloading the full block.

Schnorr Signatures: Replacing ECDSA with Schnorr signatures (as adopted in Bitcoin Taproot) would provide native non-malleability by design, signature aggregation for multi-party transactions, and improved verification efficiency.

Hybrid Consensus: Combining Proof-of-Work with Proof-of-Stake as explored in [?] would increase the economic and computational cost of 51% attacks beyond what

pure PoW achieves, improving the security model for environments with a small number of participants.

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